AN ORDINAL-PRIME TRANSFORM OF THE MÖBIUS FUNCTION

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1. Purpose and scope

This note develops the ordinal-prime transform of the Möbius function and determines exactly how much arithmetic information it carries. It is self-contained and uses no other construction.

The starting point is the Möbius function $\mu(n) \in \{-1, 0, +1\}$. The occurrences of -1 and of $+1$ are numbered separately, in increasing order of n, and the corresponding ordinal prime is attached to each occurrence; zeros are mapped to 1. The resulting signed sequence is $S(n)$ and its partial sum is $SS(n) = \sum_{k \le n} S(k)$.

The conclusions are:

(i) $SS(n)$ has an exact closed form in terms of the Mertens function $M(n)$, the squarefree counting function $Q(n)$, and the sum of the first m primes (Proposition 3.3).

(ii) After subtracting the deterministic zero contribution $n - Q(n)$, the remainder $T(n) := SS(n) - n + Q(n)$ is *exactly* a signed sum of $|M(n)|$ consecutive primes, occupying a window of prime indices symmetric about $(Q(n) + 1)/2$ (Proposition 4.1).

(iii) Consequently $T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}(1 + \theta(n))$ with $\theta(n) \to 0$ effectively, and the Mertens criterion transfers to T in both directions (Theorem 6.2).

(iv) Every object built here — SS, T, the residual E, and the centered residual E_{mid} — is a deterministic function of the pair $(M(n), Q(n))$ alone. The transform therefore creates no independent arithmetic constraint on $M(n)$. This applies to the residuals as well and is stated as Proposition 8.1.

(v) What the centered residual does isolate is a quantity whose *size* is governed by prime-gap statistics rather than by the arithmetic of μ. Section 10 reports a measured scaling

Date: Version 2.

law, $|E_{\mathrm{mid}}(n)| \asymp |M(n)|^{3/2} \log p_{Q(n)/2}$, verified over $n \le 10^6$, together with a heuristic that predicts it.

All numerical statements in this note were verified for $1 \le n \le 10^6$; the verification protocol and effective ranges are given in Section 10.

2. Notation and elementary lemmas

Let $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$ denote the primes in increasing order, and set $p_0 := 0$ by convention (used only to make the product forms below well defined at $n = 1$).

Write

$$M(n) := \sum_{k \le n} \mu(k), \qquad Q(n) := \sum_{k \le n} \mu(k)^2, \qquad \mathcal{P}(m) := \sum_{j=1}^{m} p_j, \quad \mathcal{P}(0) := 0.$$

Define the cumulative occurrence counts

$$N_-(n) = \sum_{k \le n} \frac{\mu(k)^2 - \mu(k)}{2}, \qquad N_0(n) = \sum_{k \le n} \big(1 - \mu(k)^2\big), \qquad N_+(n) = \sum_{k \le n} \frac{\mu(k)^2 + \mu(k)}{2}.$$

Thus $N_-(n)$, $N_0(n)$, $N_+(n)$ count the occurrences of $\mu = -1$, $\mu = 0$, $\mu = +1$ up to n, and the counts are inclusive of n itself.

Lemma 2.1 (Counter identities)**.** *For every $n \ge 1$,*

$$N_+(n) + N_-(n) = Q(n), \qquad N_+(n) - N_-(n) = M(n), \qquad N_0(n) = n - Q(n),$$

hence

$$N_+(n) = \frac{Q(n) + M(n)}{2}, \qquad N_-(n) = \frac{Q(n) - M(n)}{2}.$$

Lemma 2.2 (Parity)**.** $Q(n) \equiv M(n) \pmod 2$ *for every $n \ge 1$. Consequently the indices $(Q(n) \pm M(n))/2$ occurring throughout are integers.*

Proof. $Q = N_+ + N_-$ and $M = N_+ - N_-$, so $Q - M = 2N_-$. □

Lemma 2.2 is used silently in every subsequent formula and was absent from version 1; without it the central identities are not well formed.

3. The transform

Definition 3.1. The three rows are

$$R_-(n) = -\frac{\mu(n)^2 - \mu(n)}{2}\, p_{N_-(n)}, \qquad R_0(n) = 1 - \mu(n)^2, \qquad R_+(n) = \frac{\mu(n)^2 + \mu(n)}{2}\, p_{N_+(n)},$$

with the convention $p_0 = 0$ (needed only at $n = 1$, where $N_-(1) = 0$ and the coefficient vanishes). The ordinal-prime transform is

$$S(n) := R_-(n) + R_0(n) + R_+(n) = \begin{cases} -p_{N_-(n)}, & \mu(n) = -1, \\ 1, & \mu(n) = 0, \\ +p_{N_+(n)}, & \mu(n) = +1. \end{cases}$$

Along the occurrences of $\mu = -1$ the values are $-2, -3, -5, -7, -11, \ldots$; along the occurrences of $\mu = +1$ they are $2, 3, 5, 7, 11, \ldots$, in a separate ordinal count. Explicitly,

$$S(1), \ldots, S(12) = 2,\ -2,\ -3,\ 1,\ -5,\ 3,\ -7,\ 1,\ 1,\ 5,\ -11,\ 1.$$

Definition 3.2. $SS(n) := \sum_{k=1}^{n} S(k)$ and $T(n) := SS(n) - \big(n - Q(n)\big)$.

Proposition 3.3 (Closed form)**.** *For every $n \ge 1$,*

$$\boxed{SS(n) = \mathcal{P}\big(N_+(n)\big) - \mathcal{P}\big(N_-(n)\big) + N_0(n) = \mathcal{P}\left(\frac{Q(n) + M(n)}{2}\right) - \mathcal{P}\left(\frac{Q(n) - M(n)}{2}\right) + n - Q(n).}$$

Proof. The $k \le n$ with $\mu(k) = +1$ receive, in order, the primes $p_1, \dots, p_{N_+(n)}$, contributing $\mathcal{P}(N_+(n))$; those with $\mu(k) = -1$ contribute $-\mathcal{P}(N_-(n))$; each of the $N_0(n)$ zeros contributes 1. The second form is Lemma 2.1, the indices being integers by Lemma 2.2. □

4. The signed prime block

Proposition 4.1 (Block representation)**.** *Write $W(n)$ for the set of prime indices strictly between $N_-(n)$ and $N_+(n)$, inclusive of the larger endpoint; $W(n)$ has exactly $|M(n)|$ elements and is symmetric about the index $(Q(n)+1)/2$. Then*

$$T(n) = \operatorname{sgn}\big(M(n)\big) \sum_{j \in W(n)} p_j = \begin{cases} \displaystyle\sum_{j=(Q-M)/2+1}^{(Q+M)/2} p_j, & M(n) > 0, \\ 0, & M(n) = 0, \\ \displaystyle -\sum_{j=(Q-|M|)/2+1}^{(Q+|M|)/2} p_j, & M(n) < 0. \end{cases}$$

In particular

$$\operatorname{sgn} T(n) = \operatorname{sgn} M(n), \qquad T(n) = 0 \iff M(n) = 0.$$

Proof. Immediate from Proposition 3.3: $T = \mathcal{P}(N_+) - \mathcal{P}(N_-)$ is a telescoping difference of prefix sums of the primes. The window endpoints are $\min(N_\pm) + 1$ and $\max(N_\pm)$, whose midpoint is $(N_+ + N_- + 1)/2 = (Q+1)/2$. □

Corollary 4.2 (Exact bounds)**.**

$$|M(n)|\, p_{\frac{Q(n)-|M(n)|}{2}+1} \;\le\; |T(n)| \;\le\; |M(n)|\, p_{\frac{Q(n)+|M(n)|}{2}}.$$

Both inequalities are trivial (each of the $|M(n)|$ primes in the block lies between the smallest and the largest one), but the *lower* bound is what makes the transfer of Mertens bounds an equivalence rather than an implication; see Theorem 6.2.

5. The prime scale, with explicit error

We record the multiplicative estimate in a form that is valid at every n, including the infinitely many n with $M(n) = 0$. (Version 1 wrote $T(n) \sim M(n) p_{Q(n)/2}$; that assertion is empty on the zero set of M, which contains 5361 values of $n \le 10^6$.)

Proposition 5.1. *Define $\theta(n) := 0$ if $M(n) = 0$ and otherwise by $T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}\big(1 + \theta(n)\big)$. Then*

$$|\theta(n)| \;\le\; \max_{|j - Q(n)/2| \le |M(n)|/2+1} \left| \frac{p_j}{p_{\lfloor Q(n)/2 \rfloor}} - 1 \right|,$$

and $\theta(n) \to 0$ as $n \to \infty$.

Proof. The bound is Proposition 4.1 together with the triangle inequality. For the limit: by the prime number theorem in the form $M(n) = o(n)$, and by $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$, one has $|M(n)|/Q(n) \to 0$; hence the window endpoints are $(1 + o(1))Q(n)/2$ on the index scale, and $p_{(1+o(1))m}/p_m \to 1$ by the prime number theorem again. □

Both hypotheses invoked here are named: $M(n) = o(n)$ is equivalent to the prime number theorem, and $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$ is elementary.

Remark 5.2 (The asymptotic constant is a poor numerical guide)**.** From $p_m = m(\log m + \log\log m - 1 + o(1))$ and $Q(n)/2 \sim \frac{3}{\pi^2} n$ one gets

$$p_{\lfloor Q(n)/2 \rfloor} = \frac{3}{\pi^2}\, n \log n \left(1 + \frac{\log\log n - 1 + \log(3/\pi^2)}{\log n} + o\Big(\tfrac{1}{\log n}\Big) \right),$$

so the correction to the leading constant decays only like $1/\log n$. Measured values of $T(n)/\big(\frac{3}{\pi^2} n \log n\, M(n)\big)$ are 0.9558, 0.9954, 1.0151, 1.0281 at $n = 10^3, 10^4, 10^5, 10^6$ — not monotone and still 2.8% off at $n = 10^6$. By contrast $T(n)/\big(M(n) p_{\lfloor Q(n)/2 \rfloor}\big)$ equals 1.00200, 1.00071, 1.000097, 0.999998 at the same points. Any quantitative use should keep $p_{\lfloor Q(n)/2 \rfloor}$ and not the closed constant.

6. Transfer of the Mertens criterion

Proposition 6.1. *$|T(n)| = O\big(n \log n\, |M(n)|\big)$ and hence $SS(n) = O\big(n \log n\, |M(n)| + n\big)$. Unconditionally $SS(n) = o(n^2 \log n)$, by $M(n) = o(n)$ (prime number theorem).*

Theorem 6.2. *The following are equivalent.*

(a) The Riemann hypothesis.
(b) $M(n) = O_\varepsilon(n^{1/2+\varepsilon})$ for every $\varepsilon > 0$.
(c) $SS(n) - n + Q(n) = O_\varepsilon(n^{3/2+\varepsilon})$ for every $\varepsilon > 0$.

Proof. (a) $\iff$ (b) is classical (Littlewood).

(b)$\Rightarrow$(c): by Corollary 4.2, $|T(n)| \le |M(n)| p_{(Q+|M|)/2} \ll |M(n)|\, n \log n \ll_\varepsilon n^{3/2+\varepsilon} \log n$, and the logarithm is absorbed into n^ε.

(c)$\Rightarrow$(b): this is the direction version 1 asserted without proof. By the prime number theorem $|M(n)| = o(n)$ and $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$, so

$$\frac{Q(n) - |M(n)|}{2} + 1 = \big(1 + o(1)\big) \frac{3}{\pi^2} n,$$

whence the smallest prime of the block satisfies $p_{(Q-|M|)/2+1} \ge c\, n \log n$ for some $c > 0$ and all large n. The lower bound of Corollary 4.2 then gives

$$|M(n)| \le \frac{|T(n)|}{c\, n \log n} \ll_\varepsilon \frac{n^{3/2+\varepsilon}}{n \log n} \ll_\varepsilon n^{1/2+\varepsilon}. \qquad \square$$

Remark 6.3. Theorem 6.2 is a reformulation, not a proof mechanism. Note also that (b) must not be strengthened to $M(n) = O(\sqrt{n})$: the Riemann hypothesis does not imply the Mertens conjecture, which is false (Odlyzko–te Riele, 1985).

7. The zeta ingredients

The two Möbius-derived ingredients have the classical Dirichlet series

$$\sum_{m \ge 1} \frac{\mu(m)}{m^s} = \frac{1}{\zeta(s)}, \qquad \sum_{m \ge 1} \frac{\mu(m)^2}{m^s} = \frac{\zeta(s)}{\zeta(2s)}, \qquad \Re s > 1,$$

and the prime weights are governed by $\log \zeta(s)$ through the prime-counting function, via the exact Abel identity

$$\mathcal{P}(m) = m\, p_m - \int_2^{p_m} \pi(t)\, dt,$$

(verified here for $m \le 100$; e.g. $\mathcal{P}(3) = 10 = 3 \cdot 5 - 5$).

Version 1 substituted Perron integrals for M and Q into the arguments of $\mathcal{P}$. That is not a functional identity and it is dropped here, for two reasons. First, $\mathcal{P}$ is defined only on nonnegative integers, so an integral placed inside its argument is notation for "the integer that this integral evaluates to", not an analytic representation. Second, at integer x the untruncated Perron integral returns $M(x) - \frac{1}{2}\mu(x)$, not $M(x)$.

The honest statement is the following. The transform composes three classical objects,

$$\frac{1}{\zeta(s)} \ \text{(signed Möbius imbalance)}, \qquad \frac{\zeta(s)}{\zeta(2s)} \ \text{(squarefree count)}, \qquad \log \zeta(s) \ \text{(prime weights)},$$

through the nonlinear expression $\mathcal{P}\big(\frac{Q+M}{2}\big) - \mathcal{P}\big(\frac{Q-M}{2}\big) + n - Q$. None of the three components is new, and the composition is not a Dirichlet series of anything.

8. Exact information content

Proposition 8.1. *For each n, $SS(n)$ is a deterministic function of the pair $(M(n), Q(n))$; and with $Q(n)$ held fixed, $T(n)$ is strictly increasing in $M(n)$. The same is true of every residual defined below, including $E(n)$ and $E_{\mathrm{mid}}(n)$.*

Proof. The first claim is Lemma 2.1 plus Proposition 3.3. For the second, increasing M by 2 at fixed Q raises N_+ by one and lowers N_- by one, changing T by $p_{N_++1} + p_{N_-} > 0$. The residuals E and E_{mid} are T minus an explicit function of (M, Q), hence also functions of (M, Q). □

Two consequences deserve to be stated plainly.

First, $SS(n)$ does not retain the internal ordering of the ± 1 values of μ below n: two Möbius sign histories with the same endpoint counts produce the same $SS(n)$. It is a nonlinear endpoint statistic.

Second — and this corrects a claim in version 1 — the residual $E(n)$ is *not* structurally distinct from $SS(n)$ in the informational sense. Its window is determined by $(M(n), Q(n))$, so the argument above applies to it verbatim. What is genuinely different about the residual is not its information content but its *size*, which turns out to be governed by prime-gap statistics; that is the subject of the next two sections.

9. The residual and its correct centering

Definition 9.1. Let

$$p^*(n) := \tfrac{1}{2}\Big(p_{\lfloor (Q(n)+1)/2 \rfloor} + p_{\lceil (Q(n)+1)/2 \rceil}\Big)$$

be the linear interpolation of the prime sequence at the exact window centre $(Q(n)+1)/2$ of Proposition 4.1. Define

$$E(n) := T(n) - M(n)\, p_{\lfloor Q(n)/2 \rfloor}, \qquad E_{\mathrm{mid}}(n) := T(n) - M(n)\, p^*(n).$$

When $Q(n)$ is odd, $(Q+1)/2$ is an integer and $p^* = p_{(Q+1)/2}$; when $Q(n)$ is even, p^* is the mean of the two central primes.

Proposition 9.2 (Centering offset)**.** *$E(n) - E_{\mathrm{mid}}(n) = M(n)\big(p^*(n) - p_{\lfloor Q(n)/2 \rfloor}\big)$, where the bracket equals one half of a prime gap when $Q(n)$ is even and one full prime gap when $Q(n)$ is odd. In particular the bracket is positive and of size $\asymp \log p^*(n)$, so E carries a deterministic component linear in $M(n)$ which E_{mid} does not.*

This identity was verified exactly for all $n \le 10^6$, and the median of $\big(p^* - p_{\lfloor Q/2 \rfloor}\big)/\log p_{\lfloor Q/2 \rfloor}$ over $10^4 \le n \le 10^6$ is 0.5318.

Proposition 9.2 explains the correlation structure of the two residuals: the offset term is a fixed positive multiple of $M(n)$, so E correlates with M while E_{mid} does not. Measured over $100 \le n \le 10^6$:

$$\mathrm{corr}(E, M) = +0.2573, \qquad \mathrm{corr}(E_{\mathrm{mid}}, M) = +0.0045.$$

E_{mid} is therefore the object to study; E is E_{mid} plus a bookkeeping artefact.

By Proposition 4.1, when $M(n) > 0$,

$$E_{\mathrm{mid}}(n) = \sum_{j \in W(n)} \big(p_j - p^*(n)\big),$$

with the sign-reversed formula for $M(n) < 0$: a sum of deviations of the primes in the window from its interpolated centre, over a symmetric window of length $|M(n)|$.

10. Numerical verification and the measured residual law

Protocol. μ was computed by a linear sieve for $n \le N = 10^6$; M, Q by prefix sums; the prime table to index 304 071 ($p = 4\,318\,813$); T by prefix sums of the primes. At $N = 10^6$: $M(N) = 212$, $Q(N) = 607\,926$, maximum required prime index 304 069.

Exact statements verified. The following hold with no exceptions for all $1 \le n \le 10^6$: Proposition 3.3 (direct summation of S against the closed form); Lemma 2.2; Proposition 4.1 and both inequalities of Corollary 4.2; $\operatorname{sgn} T = \operatorname{sgn} M$ and $T = 0 \iff M = 0$; the window formula for E_{mid}; and the identity of Proposition 9.2.

Global residual scale ($100 \le n \le 10^6$).

	E	E_{mid}
median $\lvert\cdot\rvert$	1.09×10^3	9.74×10^2
r.m.s.	5.295×10^3	5.066×10^3
max $\lvert\cdot\rvert$	6.2766×10^4	5.7654×10^4
median $\lvert\cdot\rvert/\lvert T\rvert$	1.069×10^{-5}	8.894×10^{-6}

Growth exponents, with controls. Log-log slopes of the bin medians over 30 geometric bins on $[10^2, 10^6]$:

quantity	slope
median $\lvert E\rvert$	0.682
median $\lvert E_{\mathrm{mid}}\rvert$	0.916
control: median $\lvert M\rvert$	0.488
control: median $\lvert M\rvert \log p_{Q/2}$	0.607
control: median $\lvert M\rvert^{3/2} \log p_{Q/2}/\sqrt{12}$	0.851

The controls are the point of the table. The exponent 0.916 is not an independent growth law: it is close to the exponent of $|M|^{3/2} \log p$ and far from those of $|M|$ or $|M| \log p$.

The measured law. Set $\sigma(n) := \log p_{\lfloor Q(n)/2 \rfloor}$. The quantity

$$\rho(n) := \frac{|E_{\mathrm{mid}}(n)|}{|M(n)|^{3/2}\,\sigma(n)}$$

has the following medians, conditioned on $|M(n)|$ over $n \le 10^6$:

$\lvert M\rvert \in$	$[7,12)$	$[12,19)$	$[19,31)$	$[31,51)$	$[51,84)$	$[84,138)$	$[138,225)$	$[225,400)$
median ρ	0.131	0.133	0.143	0.143	0.140	0.140	0.136	0.140
count	46 264	58 130	86 041	127 696	152 059	164 880	231 986	65 791

The median of ρ is stable near 0.14 across a factor of 50 in $|M|$. Restricted to $n \ge 10^4$ the overall median is 0.1345; the residual drift of median ρ in n over $[10^2, 10^6]$ has log-log slope 0.13, so the law is measured, not proved, and the effective range is $10^4 \le n \le 10^6$.

Heuristic 10.1 (gap random walk)**.** Let $L = |M(n)|$ and let $g_i = p_{i+1} - p_i$ be the gaps near index $c = (Q(n)+1)/2$. Expanding each $p_j - p^*$ as a signed sum of gaps and collecting terms, one obtains the exact rearrangement

$$E_{\mathrm{mid}}(n) = \operatorname{sgn}(M) \sum_{k \ge 1} \Big(\tfrac{L}{2} - k\Big)^{+} \big(g_{m+k} - g_{m-k}\big), \qquad m = \lfloor c \rfloor,$$

in which the central gap cancels identically and the weights are triangular. If the gaps are modelled as independent with variance $\operatorname{Var}(g)$, the variance of the sum is $\operatorname{Var}(g) \sum_k 2(\frac{L}{2} - k)^2 \approx \operatorname{Var}(g) L^3/12$, giving the scale

$$E_{\mathrm{mid}}(n) \approx \frac{\sqrt{\operatorname{Var}(g)}}{\sqrt{12}}\, |M(n)|^{3/2}.$$

With the Cramér value $\operatorname{Var}(g) = \sigma^2$ this predicts median $\rho = 0.6745/\sqrt{12} = 0.1947$.

Heuristic 10.1 is a model, not a theorem; the rearrangement is exact but the independence assumption is not. Its accuracy was measured. Put

$$z(n) := \frac{E_{\mathrm{mid}}(n)}{\sigma(n)\,|M(n)|^{3/2}/\sqrt{12}}.$$

Over the 883 123 values of n with $10^4 \le n \le 10^6$ and $|M(n)| \ge 12$:

$$\text{mean } z = -0.0002, \quad \text{sd } z = 0.7265, \quad \text{median } |z| = 0.4830, \quad \text{skew} = -0.019, \quad \text{excess kurtosis} = 0.164,$$

with $\Pr(|z| < 1) = 0.834$ and $\Pr(|z| < 2) = 0.993$ against the Gaussian 0.683 and 0.955. So z is centred and symmetric but is neither of unit scale nor Gaussian: it is more concentrated than normal with slightly heavy tails. The scale deficit is largely accounted for by the gap variance being below the Cramér value — measured over prime indices $[10^5, 3.04 \times 10^5]$, $\mathrm{Var}(g)/\sigma^2 = 0.687$, predicting $\text{sd } z = 0.829$ — with the remaining 12% attributable to correlations between neighbouring gaps, which were not measured here.

Largest excursions. The extreme values of E_{mid} on $n \le 10^6$ occur in tight clusters, e.g. $E_{\mathrm{mid}} = -5.7654 \times 10^4$ at $n = 924\,067$ ($M = -301$) and $+4.6576 \times 10^4$ at $n = 932\,737$ ($M = -269$). These are the n at which $|M|$ is locally largest, consistent with the $|M|^{3/2}$ law and carrying no separate information: excursions of E_{mid} are excursions of M seen through a prime-gap window.

11. What the construction does and does not deliver

It delivers an exact, elementary dictionary between SS and (M, Q), a two-sided transfer of the Mertens criterion (Theorem 6.2), and a residual whose size obeys a clean measured scaling law.

It does not deliver an independent arithmetic constraint on $M(n)$, by Proposition 8.1. For a transformed representation to become useful it would have to produce at least one of:

(i) a cancellation unavailable in the original variables;
(ii) a positivity or monotonicity principle strong enough to yield new bounds;
(iii) a spectral representation turning a Mertens bound into an operator inequality;
(iv) an inequality between independently controlled components;
(v) a mechanism by which the prime side constrains the Möbius side beyond what (M, Q) already encodes.

None of these has been extracted here. The measured law of Section 10 is of type (iv) in form only: it bounds E_{mid} in terms of $|M|$, which is the wrong direction to be useful.

12. A variant that is not an endpoint statistic (speculative)

The obstruction identified in Proposition 8.1 is caused by the use of two *separate* ordinal counters, which destroys the ordering of the ± 1 occurrences. The minimal modification that removes it is a *shared* counter:

$$\widetilde{S}(k) := \mu(k)\, p_{Q(k)}, \qquad \widetilde{SS}(n) := \sum_{k \le n} \mu(k)\, p_{Q(k)}.$$

Here the prime index depends on k rather than on the sign-specific ordinal, so $\widetilde{SS}(n)$ is not a function of $(M(n), Q(n))$: by Abel summation with $w(k) = p_{Q(k)} \asymp k \log k$,

$$\widetilde{SS}(n) = M(n)w(n) - \sum_{k<n} M(k)\big(w(k+1) - w(k)\big),$$

so it depends on the whole trajectory of M and not only on its endpoint. This is offered as a direction, not a result; it is close to the classical weighted Mertens sums $\sum_{k \le n} \mu(k) k^{\alpha}$ and may well reduce to them.

13. Novelty status

The three zeta ingredients are classical. A search did not locate the sequence $S\,(2,-2,-3,1,-5,3,-7,1,1,5,$ as a named sequence or standard transform. Absence of a search result is weak evidence and the search was not exhaustive; this is recorded as an observation about the formulation, not a claim about its consequences, which by Proposition 8.1 follow from (M,Q).

14. Summary

$$\mu \longrightarrow (N_-,N_0,N_+) \longrightarrow S \longrightarrow SS, \qquad N_\pm = \frac{Q \pm M}{2}, \quad N_0 = n - Q, \quad Q \equiv M \ (2),$$

$$\boxed{SS(n) = \mathcal{P}\Big(\frac{Q+M}{2}\Big) - \mathcal{P}\Big(\frac{Q-M}{2}\Big) + n - Q}$$

$$\boxed{T(n) = SS(n) - n + Q(n) = \operatorname{sgn}(M) \sum_{j \in W(n)} p_j, \quad |W(n)| = |M(n)|, \text{ centre } \tfrac{Q(n)+1}{2}}$$

$$\boxed{T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}\big(1 + \theta(n)\big), \quad \theta(n) \to 0}$$

$$\boxed{\text{RH} \iff T(n) = O_\varepsilon\big(n^{3/2+\varepsilon}\big) \ \ \forall \varepsilon > 0}$$

$$\boxed{E_{\text{mid}}(n) = T(n) - M(n)p^*(n) = \operatorname{sgn}(M) \sum_{j \in W(n)} \big(p_j - p^*(n)\big), \quad |E_{\text{mid}}| \approx 0.135\,|M|^{3/2} \log p^*}$$

the last line being measured over $10^4 \le n \le 10^6$, not proved.

Every quantity above is a function of $(M(n), Q(n))$. The transform is an exact, prime-amplified reparametrisation of the Mertens function; the residual E_{mid} trades the arithmetic of μ for the statistics of prime gaps in a window of length $|M(n)|$, at the cost of a factor $|M(n)|^{3/2}$, and therefore cannot improve a bound on $M(n)$.